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Annotation Reliability in Malaysian Malay-English Code-Mixed Sentiment Analysis

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Abstract

Sentiment analysis of Malaysian Malay-English code-mixed text depends on labels that reflect how bilingual users express opinions. Yet the resources reviewed here provide limited evidence that their labels are reliable. This review examines the linguistic difficulties behind annotation, compares manual and automatic labelling approaches, and assesses the reliability evidence reported for four resources. It distinguishes neutral labels that result from missed sentiment cues from genuine differences in interpretation. MESocSentiment illustrates the concern: 78.23% of its posts are labelled neutral, and its reported error examples include Malay expressions whose positive sentiment was missed. These examples warrant further validation but do not establish the corpus-wide error rate. The review also explains why class imbalance complicates the interpretation of Cohen's kappa. Future work should report agreement for the intended task, preserve individual judgements, develop guidance for Malaysian discourse particles such as lah, and test whether automatic annotators share errors. Improving label quality and documenting its limits are necessary steps towards more dependable training and evaluation.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion across domains such as consumer services, healthcare, elections and financial services (Nazir et al., 2022). However, many language resources used for sentiment analysis are developed primarily for individual languages (Joshi et al., 2020). This creates difficulties when two languages alternate within the same message, a pattern known as code-switching. Sentiment cues may then occur in either language, while resources designed for monolingual text may not represent both languages well. Code-switched text therefore often produces lower performance than comparable monolingual text, although the size of this gap varies by task (Doğruöz et al., 2021; Winata et al., 2023).

Malaysia is one setting in which this occurs frequently: Malay-English code-mixing is common in Malaysian social media and written communication (AlAfnan

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et al., 2025). This review uses Manglish for the Malay-English combination and distinguishes it from broader Bahasa Rojak, which may also include Chinese (Romadhona et al., 2022). Treffers-Daller et al. (2022) identify recurring syntactic patterns in Malay-English switching, and Maskat and Rahman (2020) document recurring non-standard spellings such as *x* for *tidak*. This suggests that the mixing is systematic rather than random. Forms of this kind can differ from those represented in standard language resources, and may therefore obscure the words, negation or other cues that carry sentiment (Abu Bakar et al., 2020).

This mismatch between writing conventions and tool design raises the question of where the difficulty lies. One answer points to the models. Fine-tuning on code-switched data can improve results (Khanuja et al., 2020), but robustness varies across language combinations and pre-training conditions (Adilazuarda et al., 2022). Multilingual pre-trained models also do not consistently produce strong representations of code-switched text (Winata et al., 2021), and increasing multilingual coverage can introduce trade-offs in model capacity (Conneau et al., 2020). Doğruöz et al. (2021) add that widely used multilingual models such as mBERT and XLM-R are trained mainly on monolingual corpora, so they may see little or no code-switched text during pre-training. Model choice therefore explains part of the difficulty, but not all of it.

Another problem arises earlier, when sentiment labels are created. Supervised sentiment analysis depends on labelled examples for both training and evaluation (Poria et al., 2023). These labels may be assigned by humans or automatic methods, but in either case they become the supervision signal used during training and the reference against which predictions are evaluated. Incorrect training labels can teach a model patterns associated with the wrong class (Frénay and Verleysen, 2014). Incorrect test labels can distort measured performance, and can even change the ranking of competing models (Northcutt et al., 2021). Annotation reliability therefore affects how far benchmark results can be trusted (Bowman and Dahl, 2021).

For Malaysian Malay-English sentiment data, three findings make annotation reliability a priority. First, this review identifies four Malay-English sentiment resources: MY-EN-CS, SentiBahasaRojak, MyCovid-Senti and MESocSentiment (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). A usable general-purpose resource would need to meet four criteria: a scale of at least ten thousand labelled messages; a label distribution not dominated by one class; everyday social-media text; and direct evidence of annotation reliability. Each of the four satisfies some of these criteria but not all of them.

Second, MESocSentiment, the largest of the four resources, is labelled 78.23% neutral, and its published examples include Malay expressions whose positive sentiment was missed by the automatic labelling tools (Mohd Shamsuddin et al., 2024). These examples justify further validation but do not establish the corpus-wide error rate. Third, two of the four studies report no inter-annotator agreement coefficient (Kong et al., 2023; Mohd Shamsuddin et al., 2024). The other two report agreement for binary tasks, which does not establish reliability for positive, negative and neutral labelling (Kasmuri and Basiron, 2019; Romadhona et al., 2022).

Reliability is commonly assessed by asking independent annotators to label the same items and calculating a chance-corrected agreement measure such as Cohen's kappa (κ) (Cohen, 1960; Artstein and Poesio, 2008). However, strong class imbalance can make κ difficult to interpret because the expected chance agreement changes with the annotators' label distributions (Feinstein and Cicchetti, 1990; Byrt et al., 1993; Di Eugenio and Glass, 2004). A neutral-heavy corpus therefore needs both label validation and careful agreement analysis. These concerns add to the shortage of annotated resources for code-mixed Malay reported by Selvaraju et al. (2024).

These findings lead to the central question of this review: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. The review focuses on linguistic, labelling and evaluation issues only where they affect that question. It does not compare sentiment classifiers or survey sentiment analysis as a whole, topics already reviewed elsewhere (Poria et al., 2023; Nazir et al., 2022). Nor does it propose or evaluate a new system. Its contribution is to explain why the reliability question remains open and to identify priorities for future work.

This review makes three contributions:

1. It compares manual, existing-tool and generative-model annotation in terms of human effort and reliability evidence.
2. It distinguishes missed sentiment cues from genuine differences in interpretation and assesses the reliability evidence reported for four Malay-English sentiment resources.
3. It explains how linguistic difficulties, labelling choices and class imbalance interact, and identifies priorities for more reliable annotation and evaluation.

Section 2 describes how the literature was selected. Section 3 examines linguistic challenges for annotation. Section 4 compares labelling approaches and their limitations. Section 5 assesses reliability evidence. Section 6 explains how these challenges interact and proposes research priorities, and Section 7 concludes.

2. Review Methodology

The studies reviewed here were found through structured searches of the ACL Anthology, IEEE Xplore and Scopus. Each search combined three groups of terms. The first group named the language variety, and included code-mixed, code-switching, Manglish, Bahasa Rojak and Malay-English. The second group named the task, and included sentiment analysis, polarity and opinion mining. The third group named the object of interest, and included corpus, annotation, inter-annotator agreement and label quality.

The search covered work published between 2019 and 2026. Earlier work was added by following citations from the papers retrieved. The date limit was not applied to work on inter-annotator agreement, because the agreement measures discussed in this review were published before 2019.

Four criteria were used to determine inclusion. A study had to (1) address Malaysian Malay-English code-mixed text, or provide annotation or reliability methods that can be applied to it; (2) describe how its labels were produced and checked; (3) appear in a peer-reviewed journal or conference; and (4) include an annotation or reliability component. Preprints without a peer-reviewed version were not used.

The fourth criterion sets the scope of this review. Studies that only measure classifier accuracy on an existing corpus were excluded. Those studies treat the existing labels as correct, while this review asks whether the labels can be shown to be reliable.

3. Linguistic Challenges for Annotation

Reliable annotation requires attention to three features of Malay-English text: language switching, context-dependent particles and spelling variation.

3.1 Sentiment across language boundaries

A sentiment label must account for the whole message, even when its words come from different languages. Treffers-Daller et al. (2022) found frequent switching within clauses in Malaysian classroom speech. Their findings explain how Malay and English can combine within a sentence, although they do not establish how often this occurs on social media.

For annotation, the important question is whether the reader or tool recognises the words that express the opinion. A tool may understand the English part of a post but miss sentiment expressed in Malay. Fluent bilingual annotators can interpret both, but still need guidance on which opinion or target determines the label when a message contains several (Mohammad, 2016).

3.2 Particles and context

Particles such as *lah* can alter a message's tone without changing its factual content. Treffers-Daller et al. (2022) describe uses ranging from assertiveness to encouragement. AlAfnan et al. (2025) give "Don't disturb la, he's busy" as an example of a softened instruction. These studies draw on classroom speech and television dialogue, so they establish possible functions rather than their frequency in social media writing.

A particle does not have one fixed sentiment label. The instruction above might sound irritated or polite, depending on context. Guidelines therefore need to distinguish tone from positive or negative sentiment. Assigning *lah* a fixed polarity would remove the contextual distinction that makes it useful to speakers.

3.3 Spelling variation

Non-standard spelling can hide sentiment cues from automatic tools. Maskat and Rahman (2020) document forms such as *jgn* for *jangan* and several spellings of

macam mana. Their spelling-normalisation experiment achieved a highest precision of 0.43: fewer than half of the proposed matches were correct. They removed English words and excluded code-mixed posts from the evaluation subsets, so this result cannot be treated as performance on Malay-English text.

The practical concern is especially clear for negation. As discussed in Section 1, *x* can represent *tidak*. Missing that form can change how an opinion is interpreted. Normalisation should therefore be checked for its effect on sentiment, rather than judged only by whether it produces standard-looking words.

4. Annotation Approaches and Their Limitations

4.1 Human and automatic labelling

These linguistic challenges affect both human and automatic labelling. Manual annotation allows readers to interpret local language use, but requires time and relevant language skills. Kasmuri and Basiron (2019) developed a subjectivity corpus from blogs. Kong et al. (2023) used three independent annotators and a written guideline to label 11,568 tweets. Human involvement makes contextual judgement possible; it does not by itself establish reliability.

Human and automatic methods can also appear within one resource. SentiBahasaRojak combines automatically generated code-mixed reviews with stock-market posts labelled by five experts (Romadhona et al., 2022). Its reported agreement applies to the stock subset. This makes it important to describe how each part of a dataset was produced.

Existing sentiment tools can assign labels automatically. Mohd Shamsuddin et al. (2024) used TextBlob, VADER and pysentimiento to label MESocSentiment by majority vote. Although the study calls the construction process semi-automatic, its sentiment labels were assigned automatically. Whether those labels capture Malay-English sentiment must be checked separately.

Generative language models can also assign labels, but their ability to handle informal language needs task-specific validation. Jadhav et al. (2025) found that GPT-4o and Llama 3.1 405B underperformed a fine-tuned BERT model across four Marathi classification tasks, including sentiment analysis. This result does not establish their performance on Malaysian text. No study examined in this review directly evaluates generative annotation for Malay-English sentiment.

4.2 Interpreting neutral labels

MESocSentiment contains 19,714 posts, of which 78.23% are labelled neutral, 16.33% positive and 5.44% negative (Mohd Shamsuddin et al., 2024). A large neutral class is not itself evidence of error. It may reflect the sampled topics, the annotation rules, or genuine neutrality. The concern is whether some opinions received a neutral label because the tools missed their sentiment cues.

Two observations support checking for missed cues. The corpus was collected under the #Malaysia keyword across topics that include sports, politics and the arts (Mohd Shamsuddin et al., 2024). A corpus of Malaysian COVID-19 tweets labelled

by three human annotators is 23.6% neutral (Kong et al., 2023). Neither observation shows that the neutral labels are wrong, but together they justify validation.

Lexicon-based tools can miss a sentiment cue when the expression is absent from their resources. VADER combines recognised sentiment cues into a score and assigns neutral to scores near zero (Hutto and Gilbert, 2014). If a Malay opinion contributes no recognised cue, a neutral output may reflect a coverage gap.

Mohd Shamsuddin et al. (2024) present six labelled examples, three of which misclassify positive Malay expressions as neutral. One is *memang cantik sangat*, meaning "really very beautiful". These selected cases support checking for missed Malay sentiment, but they cannot estimate the error rate across the corpus.

Disagreement can reflect plausible differences in interpretation rather than annotation error (Plank, 2022). Pavlick and Kwiatkowski (2019) show that disagreement can persist in textual inference even when more judgements are collected. Neither finding establishes that ambiguity caused the neutral skew in MESocSentiment. A neutral label should not combine "no sentiment" with "uncertain interpretation" without distinguishing the two.

4.3 Limits of majority voting

Majority voting gives every annotator the same weight (Hovy et al., 2013), so an error shared by several annotators is retained. Table 1 shows that all three MESocSentiment analysers assigned neutral to most retained posts. Of the three analysers, only pysentimiento is built on a pre-trained transformer model of the kind described by Devlin et al. (2019). It also has the highest neutral share, so being built on a pre-trained transformer model did not, in this case, produce a smaller neutral share. However, class totals alone do not establish whether the tools mislabelled the same posts. That requires comparison against labels checked independently by bilingual reviewers.

Table 1. Neutral labels in the retained MESocSentiment corpus

Analyser	Neutral posts	Share
TextBlob	14,305	72.56%
VADER	13,115	66.53%
pysentimiento	16,916	85.81%
Majority vote	15,422	78.23%

Source: Mohd Shamsuddin et al. (2024). All shares use the 19,714 retained posts as the denominator.

Mohd Shamsuddin et al. (2024) discarded 512 of 20,226 posts because each analyser assigned a different label. Removing them left the neutral count unchanged at 15,422 while reducing the total from 20,226 to 19,714, so the neutral share rose by about two percentage points. Removing disputed posts may also exclude difficult cases, but the published counts do not establish their linguistic characteristics.

Retaining their text and individual labels would allow further analysis of disagreement, following the approach advocated by Uma et al. (2021).

5. Evaluating Annotation Reliability

Shared errors limit what agreement can establish: consistent judgements do not necessarily indicate correct labels. Cohen's κ compares observed agreement between two annotators with agreement expected from their label distributions (Cohen, 1960). In computational linguistics, values above 0.8 are generally treated as good reliability, and values between 0.67 and 0.8 as allowing tentative conclusions (Artstein and Poesio, 2008). The same survey cautions that interpreting a value requires knowing the task, the annotators and the distribution of their labels.

Class imbalance can make κ difficult to interpret. Di Eugenio and Glass (2004) illustrate this with two binary annotation examples: both have 90% observed agreement, yet κ ranges from −0.048 to 0.80 because the category distributions differ. A single threshold therefore cannot be applied to every agreement result.

For a neutral-heavy dataset, researchers should examine which classes attract agreement and which attract disagreement. The final corpus distribution is a reason to investigate prevalence, but κ depends on the individual annotators' distributions. Reliability assessment remains possible; it needs more evidence than an aggregate coefficient or class count.

Table 2. Reliability evidence reported for the reviewed resources

Resource	Annotation evidence	Scope and limitation
MY-EN-CS Kasmuri and Basiron (2019)	Two bilingual undergraduates; written guideline; κ = 0.83.	Opinionated versus factual judgement. Does not establish three-class polarity agreement.
SentiBahasaRojak Romadhona et al. (2022)	Five experts labelled stock posts; reported kappa = 0.77.	Positive versus negative stock labels; coefficient variant unspecified. Other subsets were generated automatically.
MyCovid-Senti Kong et al. (2023)	Three independent annotators; written guideline.	Three-class sentiment labels; no agreement coefficient reported.
MESocSentiment Mohd Shamsuddin et al. (2024)	Majority vote among three automatic analysers.	No human inter-annotator agreement coefficient reported.

Table 2 shows gaps in the available evidence, rather than proving that all four resources are unreliable. Two studies report no coefficient; the other two report binary-task agreement. Those binary results are informative for their stated tasks, but cannot be transferred to three-class sentiment annotation. Romadhona et al. (2022), for example, used five experts with the final label set by majority vote, whereas Cohen's κ is defined for two annotators. Studies with more than two annotators should therefore name the coefficient used and explain how it was calculated.

6. Research Priorities

The reviewed studies leave uncertainty about how labels were produced and checked. Section 6.1 first sets out how the challenges described in Sections 3 to 5 interact. Four research priorities then follow: annotation reporting, agreement under class imbalance, guidance for particles and ambiguity, and testing for shared errors.

6.1 How the challenges interact

The three challenges examined above are connected. Particles and spelling variation mean that some sentiment cues can be read in more than one way or missed altogether. Reliable labelling therefore needs fluent bilingual readers, and their cost is the reason Mohd Shamsuddin et al. (2024) give for automating. The tools used for automation were designed for other languages (Hutto and Gilbert, 2014). Where none of them recognises a Malay sentiment cue, majority voting confirms the missed cue rather than correcting it, and the neutral class grows.

That skew then affects the statistic used to check the labels. A label distribution dominated by one class is the condition under which Cohen's κ becomes difficult to interpret (Feinstein and Cicchetti, 1990; Byrt et al., 1993). A researcher who suspects the labels and computes κ therefore cannot easily distinguish poor annotation from competent annotation of a skewed corpus. Corpora whose reliability was never established are then released and used to train and evaluate models. A model trained on such labels learns the patterns associated with them (Frénay and Verleysen, 2014). If that model is later used as an annotator, the sequence repeats.

This sequence helps explain why improvements at individual stages have not yet produced a resource whose reliability can be established. Each reviewed study addresses one stage, but an intervention at one stage is assessed with a statistic that another stage has made difficult to interpret. The four priorities below each address one point in the sequence.

6.2 Report how labels were produced

Corpus releases should state who or what assigned the labels, provide the guideline, and identify the task and agreement measure. Data statements can document the populations and conditions to which a resource applies (Bender and Friedman, 2018). Releases should also report class distributions, agreement by class, individual judgements and discarded items. Preserving individual annotations supports methods that model differences between annotators instead of reducing them to a majority label (Mostafazadeh Davani et al., 2022).

6.3 Evaluate agreement under class imbalance

Researchers should compare agreement measures on data that retain independent annotations, examining how results change as class proportions vary. One option is prevalence adjustment, which changes the chance-agreement calculation to reduce the effect of unequal category frequencies (Byrt et al., 1993). Its assumptions and suitability for three-class sentiment need evaluation; adjustment does not

automatically make a measure appropriate (Di Eugenio and Glass, 2004). Corpora containing only final labels would require additional annotation.

6.4 Develop guidance for particles and ambiguity

Annotation guidelines should show how context changes the interpretation of particles such as *lah* and *kan*. Mohammad (2016) separates questions about the speaker, opinion target and attitude, and Kong et al. (2023) adapt this approach for Malaysian tweets. A useful extension would include local particle examples and a clear rule for unresolved ambiguity. Shared guidance can reduce avoidable disagreement while preserving readings that remain genuinely uncertain.

6.5 Test whether annotators share errors

Bilingual humans, existing sentiment tools and generative models should label the same Malay-English sample. A separate group of bilingual reviewers could assign reference labels independently, explain their decisions and discuss disagreements. Unresolved cases should retain the competing labels. Human review of this kind can also improve the automatic detection of annotation errors (Weber and Plank, 2023). Weber-Genzel et al. (2024) use explanations and subsequent validity judgements to distinguish annotation error from plausible variation in English textual inference. A similar procedure could support the proposed Malay-English comparison. Researchers could then test whether voting reduces errors or preserves mistakes shared by the annotators.

7. Conclusion

This review examined why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. Part of the difficulty lies in the models, but another part arises earlier, in how sentiment labels are produced and checked. The reviewed resources show incomplete agreement reporting, evidence of missed Malay sentiment cues, and uncertainty about the benefits of majority voting. These difficulties are connected. The labelling methods produce skewed class distributions, and that skew makes the agreement coefficient used to check the labels difficult to interpret. Progress therefore depends on transparent annotation reporting, guidance for context-sensitive expressions, agreement analysis that accounts for class imbalance, and direct tests of whether annotators share errors. Until reliability can be assessed under these conditions, a new corpus cannot be shown to improve on the ones examined here.

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